ME2322
Industrial transformation and technological innovation

Emrah Karakaya

2026-10-27

Few words about myself

Emrah Karakaya, Associate Professor (Docent)

Theme Leader at KTH Food Centre & **Faculty member** at Digital Futures

Two ongoing research projects

- AI and Industrial Transformation (WASP-HS, Wallenberg Foundations)
- Data-Driven Sustainable Food (KTH Digital Futures)

Teaching: ME2322; Sustainable Business; Organizational Implications of AI



Previous research : Transitions in energy, aviation, steelmaking, mobility etc.

International exposure: Türkiye, Spain, Germany, the USA, and Sweden

Extensive research experience: 18 papers, 2500+ citations etc.

https://www.kth.se/profile/emrahka

What do you expect to learn in this course?

- First, talk to your peers



- Then, answer mentimeter
  - Join at menti.com
    - Use the code XXXX XXXX

ME2322 – Intended learning outcomes

- Describe, explain and **use concepts, models and theories**
  - to analyze industrial dynamics and technological innovation.
    - among others applied to **industrial development in Sweden**

- Analyze how **societal challenges** affect industrial dynamics and technological innovation, as well as
  - how industrial dynamics and technological innovation affect the **societal response to these challenges**.

Navigating the emerging industrial transitions

NEWSLETTERS· CFO DAILY

MIT report: 95% of generative AI pilots at companies are failing

BY SHERYL ESTRADA
SENIOR WRITER AND AUTHOR OF CFO DAILY

August 18, 2025 at 6:54 AM EDT

FORTUNE



GETTY IMAGES

Krisen i Stegra

”Är Stegra på väg mot en dödsfälla?”



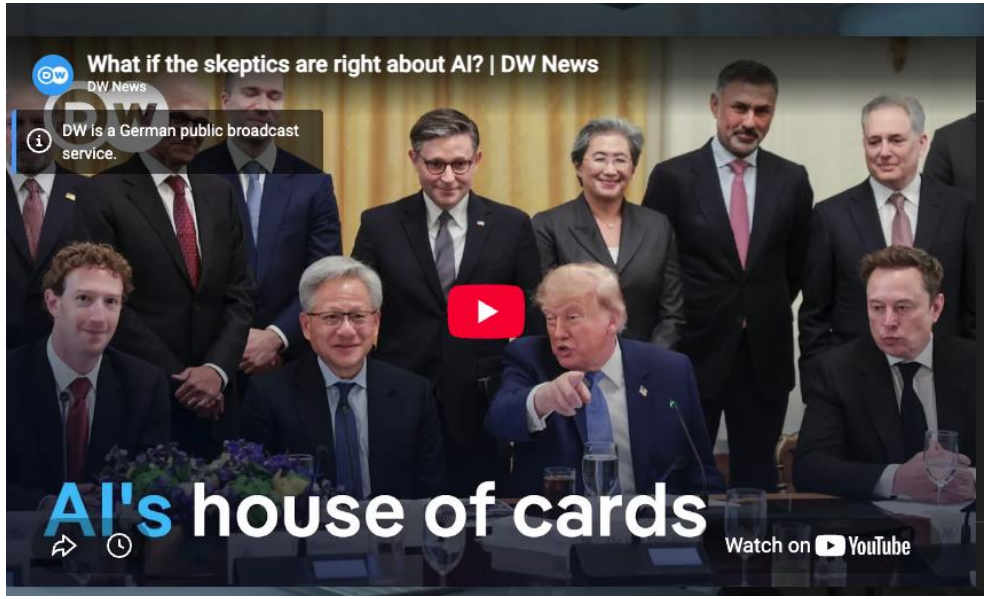
Svenska Dagbladet

Maten och klimatet

”Den planetära dieten hotar vår matförsörjning”



Svenska Dagbladet



2026-10-16

A new design of the course, set in 2025

- **Updated course literature & a new structure**

- **Theory:** Research-based knowledge
  - That covers both supply and demand sides of technological innovation

- **Practice:** Guest lectures
  - Four industries: Food, Energy, Gaming & Health
  - Two societal challenges: AI-driven Transformation & Sustainability Transitions

- **Student survey 2025**
  - Overall impression score: 4.6/5
  - *A student comment:*

    ”Overall, a very nice, small course that I learned a surprising amount from. Definitely applicable to future work.”

The structure

	Monday	Tuesday	Wednesday	Thursday	Friday
Week 44		Intro			
Week 45		L1	L2	L3	
Week 46		G1	G2		
Week 47	G3		G4	L4	
Week 48		S,S	S,S		
Week 49		Tutorial		Oral Exams	Oral Exams
Week 50	Oral exams	Oral exams	Oral exams		
Week 51					
Week 52					
Week 1					Assignment

Research-based lectures, covering *theory*

- L1, **Industrial dynamics**: dominant design and general-purpose technologies
- L2, **Socio-technical transitions**: technological innovation systems and multi-level perspective
- L3, **Diffusion of innovations**: Adopters and change agents

- L4, **Critical perspectives** on transitions towards artificial intelligence and sustainability

Guest lectures, covering *practice*

- G1: Food sector
  - One Planet Plate (Anna Wahlberg, Programme Manager Sustainable Food at WWF Sweden)



WWF | ONE PLANET PLATE 2025
En måltid för en levande planet.

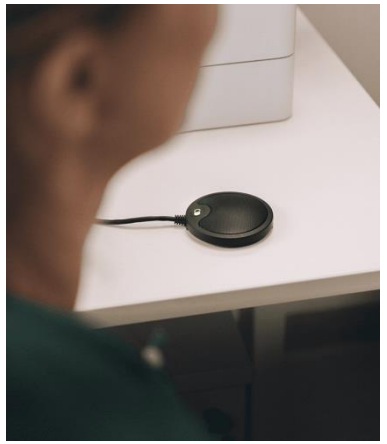
- G2: Energy sector
  - AI for renewable energy (John Diklev, Founder and CEO at Flower)



FLOWER

Guest lectures, covering *practice*

- G3: Healthcare
  - AI for clinics (Tim Gisseman, Operations Associate at Tandem)



Tandem

- G4: Gaming industry
  - The case of large-language models (Patrick Ghirmai Juhl, AI Program Director, King)

King

Theory-driven literature

- Industrial dynamics
  - Utterback, J.M., 1996. Dominant Design and the Survival of Firms - How companies can seize oppurtunities in the face of technological change, in: Utterback, J.M. (Ed.), Mastering the Dynamics of Innovation. Harvard Business School Press, Boston, Massachusetts, pp. 23–56.
  - Bresnahan, T. (2024). What innovation paths for AI to become a GPT?. Journal of Economics & Management Strategy, 33(2), 305-316. https://doi.org/10.1111/jems.12524

- Socio-technical transitions
  - Bergek, A., Jacobsson, S., Carlsson, B., Lindmark, S. and Rickne, A., 2008. Analyzing the functional dynamics of technological innovation systems: A scheme of analysis. Research policy, 37(3), pp.407-429. https://doi.org/10.1016/j.respol.2007.12.003
  - Markard, J., Raven, R. and Truffer, B., 2012. Sustainability transitions: An emerging field of research and its prospects. Research policy, 41(6), pp.955-967. https://doi.org/10.1016/j.respol.2012.02.013
  - Karakaya, E., Nuur, C., & Assbring, L. (2018). Potential transitions in the iron and steel industry in Sweden: towards a hydrogen-based future?. Journal of cleaner production, 195, 651-663. https://doi.org/10.1016/j.jclepro.2018.05.142

Theory-driven literature

- Diffusion of innovations
  - Wejnert, B. (2002). Integrating models of diffusion of innovations: A conceptual framework. Annual review of sociology, 28(1), 297-326. https://doi.org/10.1146/annurev.soc.28.110601.141051
  - Karakaya, E., & Sriwannawit Lundberg, P. (2020). International Encyclopedia of Human Geography (Second Edition), https://doi.org/10.1016/B978-0-08-102295-5.10638-9

- Critical Perspectives
  - Susur, E., & Karakaya, E. (2021). A reflexive perspective for sustainability assumptions in transition studies. Environmental Innovation and Societal Transitions, 39, 34-54. https://doi.org/10.1016/j.eist.2021.02.001
  - Bennich, A. (2025). Untangling digitalisation: a topic of growing relevance for transition scholars. Environmental Innovation and Societal Transitions, 57, 101021. https://doi.org/10.1016/j.eist.2025.101021
  - Lai, Y. Y., & Karakaya, E. (2024). Rethinking the sustainability of transitions: An illustrative case of burden-shifting and sociotechnical dynamics of aviation fuel in Sweden. Energy Research & Social Science, 113, 103574. https://doi.org/10.1016/j.erss.2024.103574

Case-based literature

- Food
  - Praasterink, F., Beers, P. J., Hassink, J., House, J., & Van Der Horst, H. (2025). Assessing ongoing sustainability transitions: The state of food system transition in the Netherlands. NJAS: Impact in Agricultural and Life Sciences, 97(1), 2534486. https://doi.org/10.1080/27685241.2025.2534486
- Energy
  - Christley, E., Lai, Y. Y., Brauer, H. B., & Ingersoll, A. A. (2025). A beginner's guide to reflexivity in energy research and social science. Energy Research & Social Science, 127, 104267. https://doi.org/10.1016/j.erss.2025.104267
- Health
  - Johansen, F., Loorbach, D., & Stoopendaal, A. (2018). Exploring a transition in Dutch healthcare. *Journal of health organization and management*, *32*(7), 875-890. https://doi.org/10.1108/JHOM-07-2018-0185
- Artificial Intelligence
  - van den Broek, E. (2025). Unpacking AI at work: Data work, knowledge work, and values work. Information and Organization, 35(3), 100584. https://doi.org/10.1016/j.infoandorg.2025.100584

Examination

- The examination of the course includes
  - TEN1: an oral exam
  - INL1: to assignments
    - an assignment connected to the seminars
    - another assignment in the format of a reflection essay in the end of the course.

TEN1 (2 ECTS)

- **An individual examination**
  - But, simultaneously taken by three students
- The students will **self-assign** themselves for provided time slots on
  - 3 December, 4 December, 7 December, 8 December or 9 December.
- **The format**
  - Intro
  - Quiz: a mix of multiple-choice and open-ended questions on paper
  - Reflection: follow-up questions for each student to reflect deeper on their answers
- **Content**
  - Theory and practice
  - **Tip: Attend the lectures! The open-ended questions will be related to what is discussed in the lectures**

INL1 (Two assigments)

- Part 1, group work
  - Each team (of 4-5 students) will choose one of the research questions provided in the course and write a short report (approx. 1000 words) and present their findings in one the seminars.
  - The research topics that students can choose will be announced soon and will target both *theory* and *practice* parts of the course.

- Part 2, individual reflection:
  - In the end of the course, each students will write a self-reflection report (approx. 500), covering their critical and reflexive learning process in the course.

Plagiarism

- In this course, plagiarism is not accepted. All the submissions will be checked for plagiarism.
- More information on plagiarism (along with a handbook) can be found at: https://www.kth.se/en/student/stod/studier/fusk-1.997287

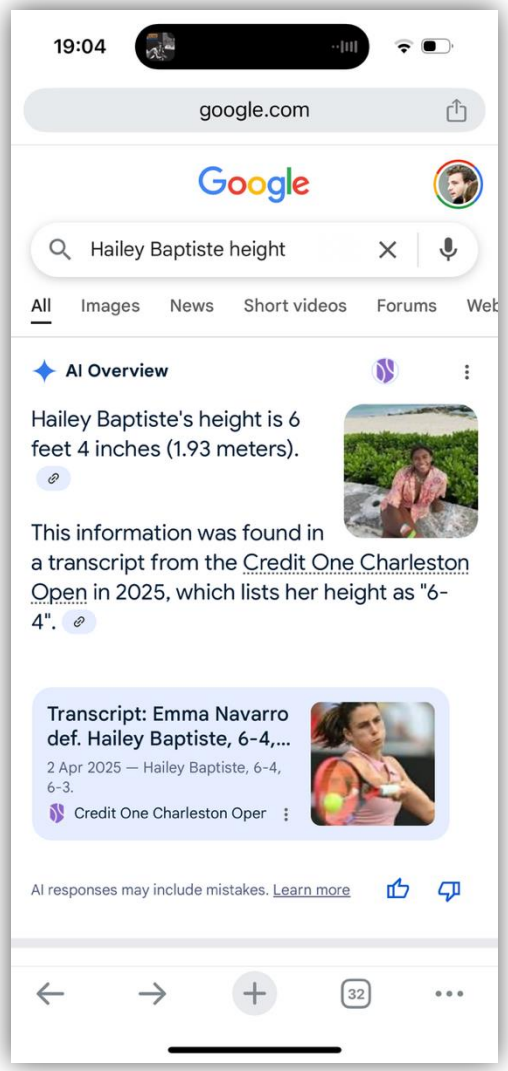
Use of Generative AI

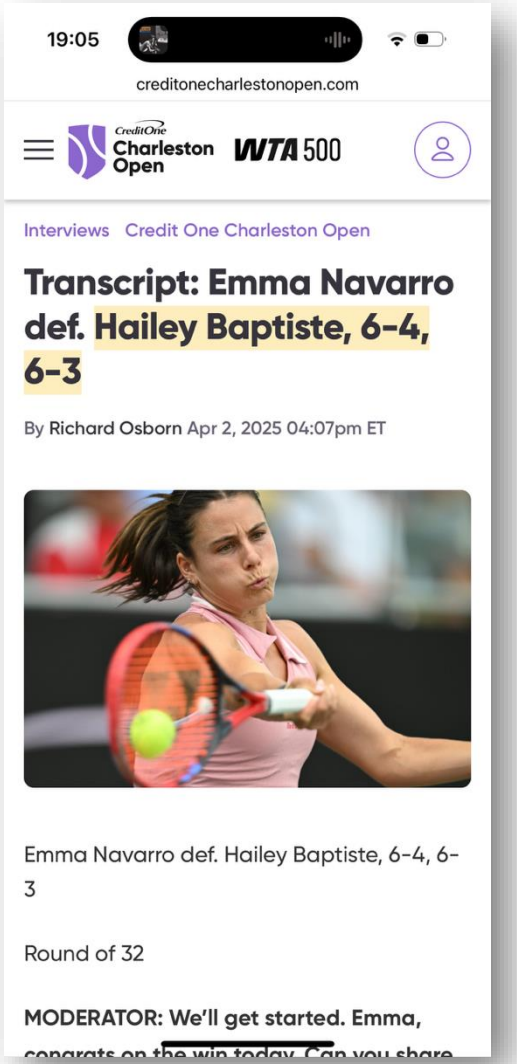
- Have you used generative AI in other courses?
- How was your experience?

Can LLMs help us to get closer to the “truth”?

What is the height of Hailey Baptiste?







Another example (new)

User profiles for **emrah karakaya**

Emrah Karakaya
KTH Royal Institute of Technology
Verified email at indek.kth.se
Cited by 2571

[HTML] A reflexive perspective for sustainability assumptions in transition studies

[HTML] sciencedirect.com

E Susur, E **Karakaya** - Environmental Innovation and Societal Transitions, 2021 - Elsevier

✦ Quick read

The sustainability transitions literature addresses globally pressing sustainability challenges. However, the scholars in the field trend towards using the term “sustainability” in a manner …

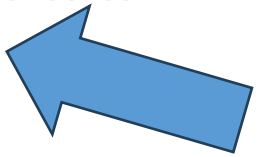
☆ Save 99 Cite Cited by 92 Related articles All 5 versions

[HTML] Experimental networks for business model innovation: A way for incumbents to navigate sustainability transitions?

[HTML] sciencedirect.com

M Engwall, M Kaulio, E **Karakaya**, M Miterev, D Berlin - Technovation, 2021 - Elsevier

✦ Quick read



To navigate sustainability transitions, firms are often prompted to take an active role in business model innovation. Previous research has shown, however, that when attempting to …

☆ Save 99 Cite Cited by 99 Related articles All 6 versions

Quick read

Experimental networks for business model innovation: A way for incumbents to navigate sustainability transitions?

M Engwall, M Kaulio, E Karakaya, M Miterev, D Berlin

Technovation 108, 102330, 2021 Cited by 99

✦ *[HTML] sciencedirect.com*

Presents a conceptualization of Experimental Networks (ENs) as a strategic vehicle for incumbent firms to pursue Business Model Innovation (BMI) aimed at sustainability, focusing on the roles, characteristics, and operational aspects of such networks.

Answers

Show details

EN conceptualization: Experimental Networks (ENs) are conceptualized as strategic vehicles for incumbent firms to pursue Business Model Innovation (BMI) aimed at sustainability, providing a way to navigate sustainability transitions by managing inherent tensions between maintaining current operations and exploring new sustainable practices.

EN roles and activities: The core activities of an Experimental Network include the Incumbent framing the problem, the Incubator coordinating and managing resources, the Complementors providing assets and knowledge, and the Users offering feedback and participating in testing.

Example EN implementation: An exemplary Experimental Network was implemented in Finland between 2018 and 2019, focused on developing new, sustainable business models for bio- and circular economy solutions in the forest industry, involving 12 participating organizations.

Approach

Conceptual Development: The approach is primarily conceptual, drawing on literature concerning business model innovation, sustainability transitions, and organizational ambidexterity to develop the Experimental Network framework.

Case Study Illustration: The conceptual framework is illustrated and partly informed by an empirical case study of an Experimental Network implemented in the Finnish forest industry between 2018 and 2019, involving 12 organizations.

OKish

Getting strange ….

Finland!?? Forest!???
OMG – completely wrong!

The Deloitte's GenAI Scandal

Deloitte to pay money back to Albanese government after using AI in $440,000 report

Partial refund to be issued after several errors were found in a report into a department's compliance framework

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Deloitte will provide a partial refund to the federal government over a $440,000 report that contained several errors, after admitting it used generative artificial intelligence to help produce it.

The Department of Employment and Workplace Relations (DEWR) confirmed Deloitte would repay the final instalment under its contract, which will be made public after the transaction is finalised. It comes as one Labor senator accused the consultancy firm of having a "human intelligence problem".

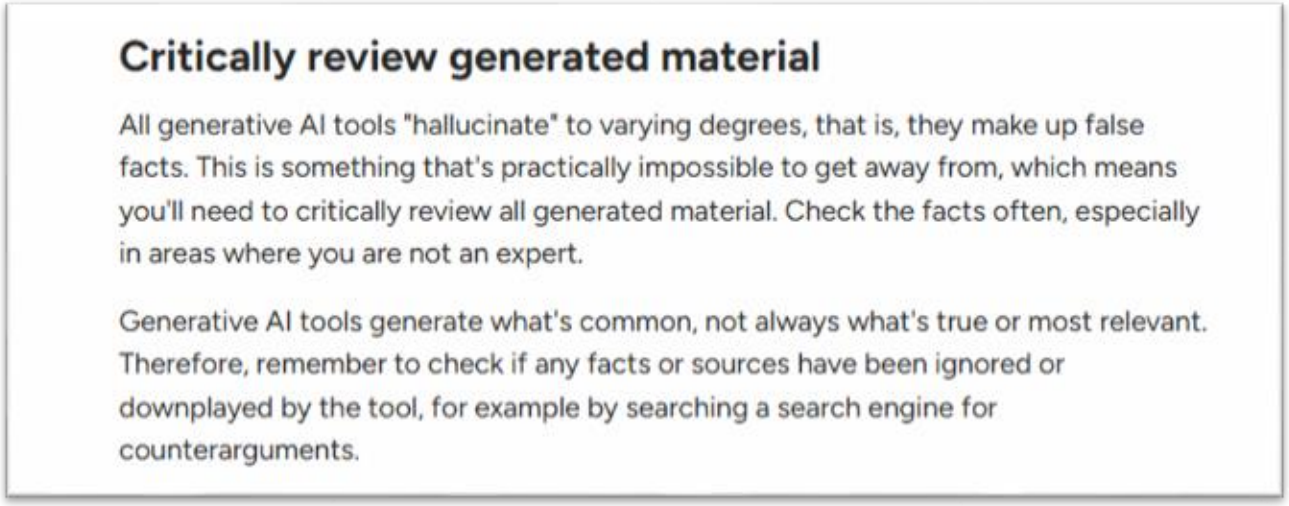
Deloitte was commissioned by the department to review the targeted compliance framework and its IT system, used to automate penalties in the welfare system if mutual obligations weren't met by jobseekers, in December 2024.

[The Guardian, 6 October 2025]

The Guardian

Course-specific genAI guidelines

- Generative AI tools allowed according to given guidelines
- In ME2322, for INL1
  - Be transparent with your use of generative AI, which means that your use must be disclosed in an Appendix (at least with half page explanation)
  - You should critically review generated material



Critically review generated material

All generative AI tools "hallucinate" to varying degrees, that is, they make up false facts. This is something that's practically impossible to get away from, which means you'll need to critically review all generated material. Check the facts often, especially in areas where you are not an expert.

Generative AI tools generate what's common, not always what's true or most relevant. Therefore, remember to check if any facts or sources have been ignored or downplayed by the tool, for example by searching a search engine for counterarguments.

https://intra.kth.se/en/utbildning/systemstod/generativ-ai/use-generative-ai-efficiently-1.1253700

A reference group ME2322

- We will form a course reference group (Emrah, Beatriz, and some student reps).
  - The purpose is to discuss ongoing activities in the course and exchange feedback.
  - Meeting two- three times at lunch times
    - Lunch provided
  - If you are interested in participating, talk/write to Emrah (emrahka@kth.se) ASAP





Any questions or comments?

Emrah Karakaya

emrahka@kth.se